MACHINING AND MACHINE TOOLS

Metal Cutting Mechanics • Tool Geometry • Chips • Cutting Forces • Tool Life • Cutting Fluids

Reference: Elements of Workshop Technology, Vol. II - Hajra Choudhury

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1. Introduction and Basic Elements of Machining

Machining is a manufacturing process in which unwanted material is removed from a workpiece in the form of chips by means of a cutting tool, so that the required shape, size, dimensional accuracy and surface finish are obtained. Turning, shaping, planing, drilling, milling, boring and grinding are common machining operations.

1.1 Basic elements of machining

Element	Function / meaning
Workpiece	The material from which excess material is removed.
Cutting tool	The wedge-shaped element that penetrates the work and causes plastic deformation and chip separation.
Machine tool	Provides controlled relative motion, rigidity and power for the cutting operation.
Cutting motion	Primary motion responsible for chip formation; e.g., rotation of work in turning.
Feed motion	Secondary motion that continuously brings fresh material into the cutting zone.
Depth of cut	Thickness of work material removed in a single pass, measured normal to the machined surface.
Cutting speed	Relative velocity between cutting edge and work in the direction of primary cutting motion.
Cutting fluid	Fluid supplied to cool, lubricate, flush chips and improve tool life/surface quality.

1.2 Common machining variables

- Cutting speed V: speed of the cutting edge relative to the work, normally expressed in m/min.
- Feed f: advance of tool per revolution, stroke or tooth, depending on the process.
- Depth of cut d: thickness of layer removed in a pass.
- Uncut chip thickness t1: chip thickness before deformation.
- Chip thickness t2: actual thickness of chip after cutting.
- Rake angle (α): controls chip flow, cutting force and strength of the cutting edge.
- Tool-work material combination: strongly affects cutting force, wear and permissible cutting speed.

2. Sources of Heat in Metal Cutting

Almost all mechanical energy supplied in machining is converted into heat. The temperature of the cutting zone is therefore governed by the rate of heat generation and the way in which heat is carried away by the chip, workpiece, tool and cutting fluid.

Heat source	Cause	Importance
Primary deformation zone	Heat generated by severe plastic deformation of the work material along and around the shear zone.	Usually the major source.
Tool-chip interface	Friction and secondary plastic deformation as the chip slides over the rake face.	Very important at medium and high cutting speeds.
Tool-work / flank interface	Friction between flank face and newly machined surface.	Increases greatly when flank wear develops.

- The chip generally carries away a large fraction of the heat at high cutting speeds.
- The tool receives a smaller fraction of total heat but can attain very high local temperatures.
- Excessive temperature accelerates diffusion, oxidation and softening of the tool.
- Cutting fluids reduce temperature and/or friction depending on their composition and method of application.

3. Orthogonal and Oblique Cutting

3.1 Orthogonal cutting

In orthogonal cutting, the cutting edge is perpendicular to the direction of cutting velocity. The chip flows approximately normal to the cutting edge, so the process can be treated as a two-dimensional cutting problem.

3.2 Oblique cutting

In oblique cutting, the cutting edge is inclined at an angle to the direction of cutting velocity. The chip flows sideways across the rake face and the mechanics are three-dimensional. Most practical turning, milling and drilling operations involve oblique cutting.

Point	Orthogonal cutting	Oblique cutting
Cutting edge relative to cutting velocity	Perpendicular	Inclined
Chip flow	Normal to cutting edge	Sideways; at a chip-flow angle
Nature of analysis	Two-dimensional	Three-dimensional
Force components	Normally resolved into cutting and thrust forces	Cutting, feed/thrust and radial components
Practical occurrence	Idealized analysis and some special operations	Most practical metal-cutting operations
Cutting action	More abrupt	Generally smoother with gradual engagement along the edge

4. Cutting Tools and Single Point Cutting Tool (SPC)

4.1 Types of cutting tools

Type	Feature	Examples
Single-point tool	One main cutting edge	Turning, shaping, planing, boring
Double-point tool	Two principal cutting edges	Twist drill
Multi-point tool	More than two cutting edges	Milling cutter, broach, reamer, grinding wheel

4.2 Parts of a single point cutting tool

Part	Function / description
Shank	Main body held in the tool holder.
Face / rake face	Surface over which the chip flows.
Side flank	Surface below the side cutting edge.
End flank	Surface below the end cutting edge.

3. Orthogonal and Oblique Cutting

Figure: Orthogonal (a) and oblique (b) cutting

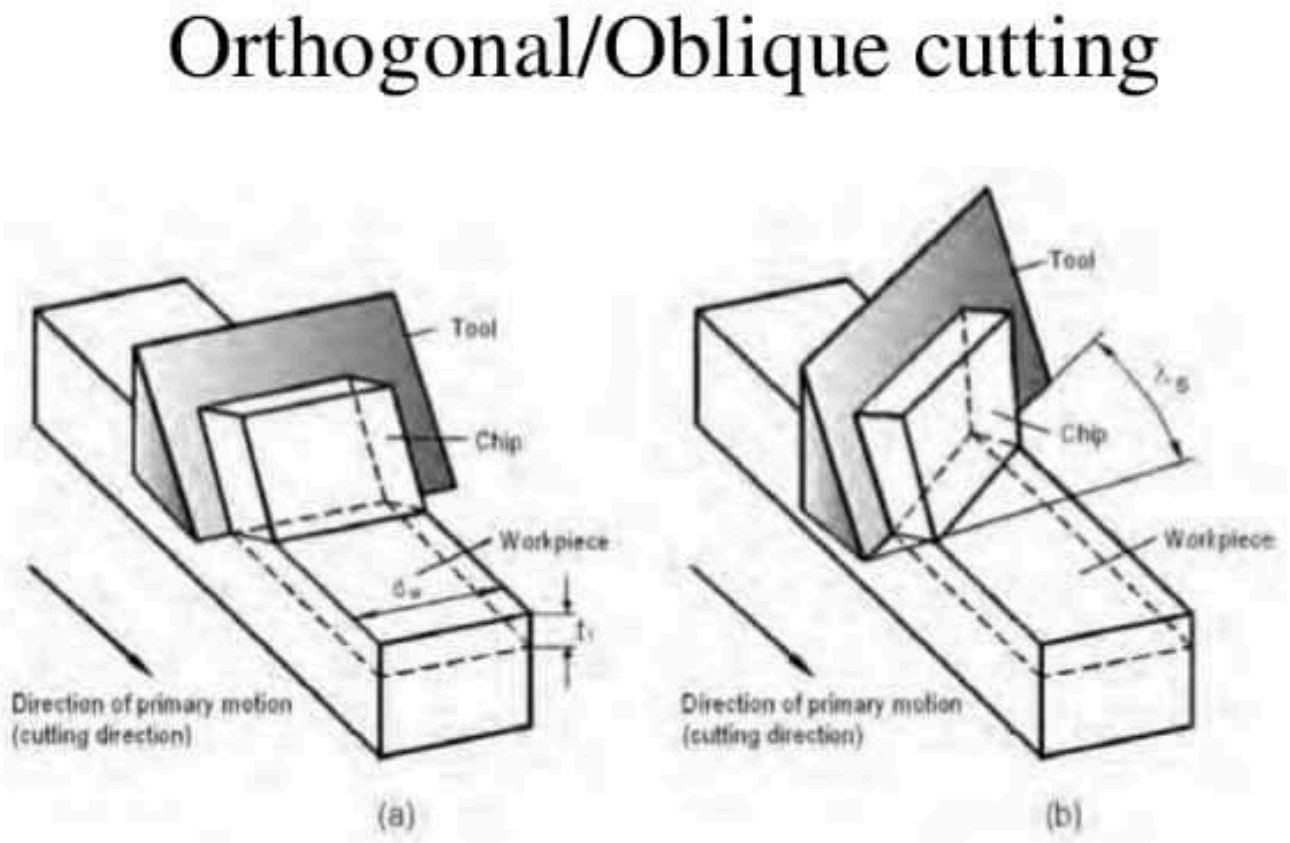


Fig. 2.3. Orthogonal (a) and oblique (b) cutting

Orthogonal cutting: the cutting edge is perpendicular to the cutting velocity.

Oblique cutting: the cutting edge is inclined, and the chip flows sideways.

Side cutting edge	Intersection of face and side flank; performs most of the cutting.
End cutting edge	Intersection of face and end flank.
Nose	Junction of side and end cutting edges.
Nose radius	Rounded portion at the nose; improves finish and edge strength.
Base	Bottom surface of the tool shank used for setting and support.

5. Principal Tool Angles, Tool Signatures and Reference Planes

5.1 Principal angles of a single point cutting tool

Angle / feature	Purpose
Back rake angle	Slope of tool face from nose towards the rear; influences chip flow and cutting force.
Side rake angle	Slope of tool face sideways; helps direct the chip and influences cutting force.
End relief angle	Clearance between end flank and machined surface; prevents rubbing.
Side relief angle	Clearance between side flank and work; prevents rubbing during feed.
End cutting edge angle	Angle locating the end cutting edge relative to a reference direction.
Side cutting edge angle	Angle locating the side cutting edge; affects chip thickness, force distribution and tool engagement.
Nose radius	Radius joining side and end cutting edges; affects finish and tool strength.

Principal angles of a single-point cutting tool

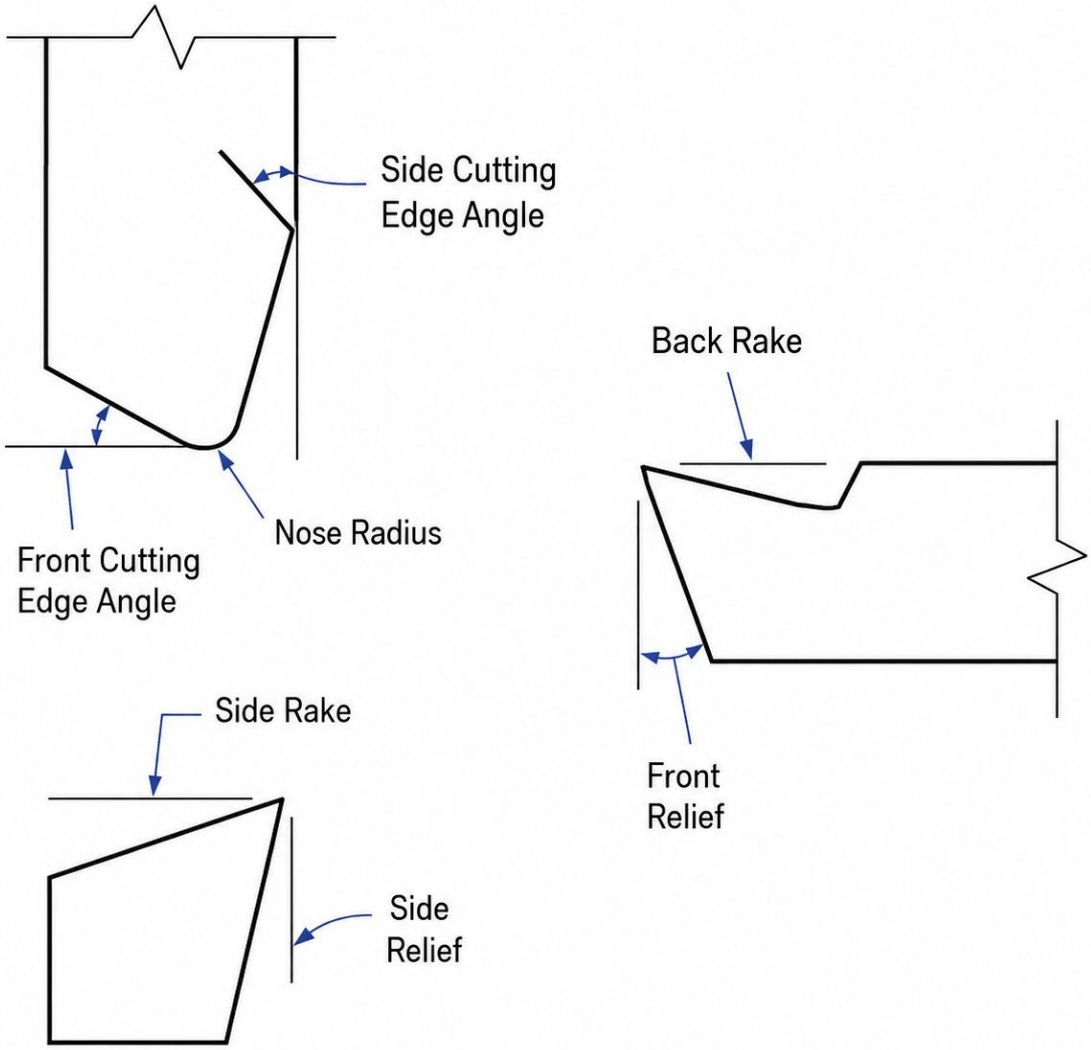


Figure: Side cutting-edge angle, front cutting-edge angle, nose radius, side rake, back rake, side relief and front relief.

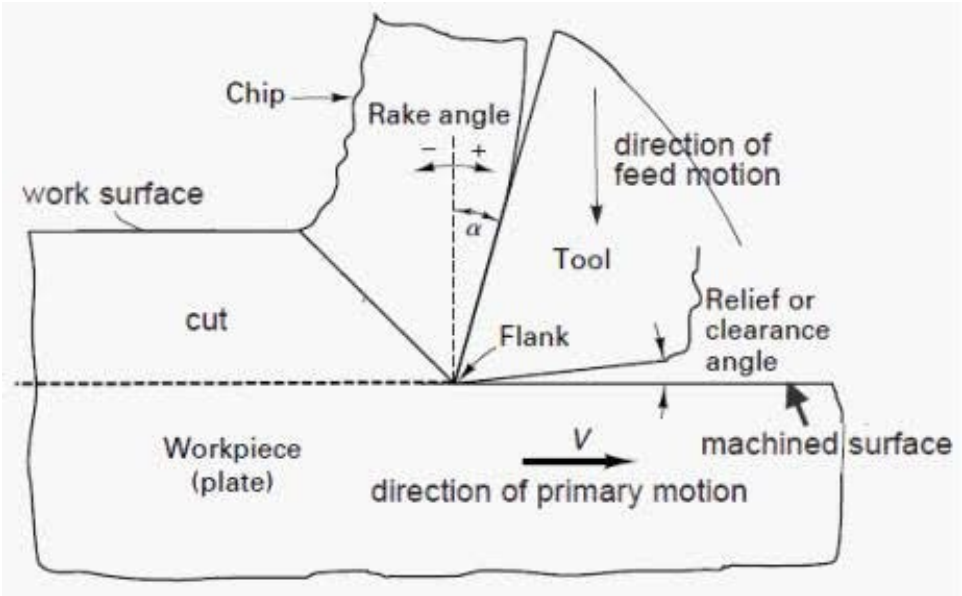


Figure 2. Rake angle α and relief/clearance angle at the cutting edge during machining.

5.2 Tool signature in ASA system

The American Standards Association (ASA) system specifies the tool geometry in the order:

Back rake - Side rake - End relief - Side relief - End cutting edge angle - Side cutting edge angle - Nose radius (inch)

Example: 8 - 10 - 6 - 6 - 8 - 15 - 1 mm represents the corresponding seven tool parameters in the above sequence.

5.3 Tool signature in ORS system

In the Orthogonal Rake System (ORS), tool angles are referred to the orthogonal plane. A commonly used ORS tool signature is written as:

Inclination angle - Orthogonal rake angle - Orthogonal side relief angle -
Orthogonal end relief angle - End cutting edge angle - Approach angle - Nose radius (mm)

The ORS is especially useful in metal-cutting analysis because the rake and clearance angles are measured in the orthogonal plane.

5.5 Reference planes used in tool geometry

Plane	Definition / purpose
Reference plane	Plane passing through the selected point on the cutting edge and normal to the cutting velocity.
Cutting plane	Plane tangent to the cutting edge at the selected point and containing the cutting velocity direction.
Orthogonal plane	Plane perpendicular to both the reference plane and cutting plane; used for orthogonal rake and clearance angles.
Normal plane	Plane normal to the cutting edge at the selected point.
Machine reference plane	Reference plane connected to machine-tool axes; useful for tool setting and grinding.

6. Chip Formation and Types of Chips

6.1 Chip formation mechanism

When the cutting tool penetrates the work, material ahead of the tool is compressed. As stress rises beyond the flow strength, intense plastic deformation occurs in a narrow region. The deformed layer then moves upward along the rake face as a chip. In the simplified shear-plane model, the principal deformation is treated as occurring along a shear plane inclined at shear angle φ.

6.2 Types of chips

Chip type	Typical conditions	Effects
Continuous chip	Ductile work material, high cutting speed, sharp tool, large positive rake, good lubrication.	Good surface finish; long chip may be hazardous.
Continuous chip with built-up edge (BUE)	Ductile material at low/medium speed, high friction, small rake, poor lubrication.	Poor and varying finish; cutting force fluctuates; tool may be protected temporarily by BUE.
Discontinuous chip	Brittle material, low speed, small/negative rake, vibration or high friction.	Short segments; fluctuating force; surface may be rough.
Serrated / segmented chip	High-speed cutting of difficult alloys such as titanium/nickel alloys; localized shear.	Saw-tooth chip; cyclic force and temperature variations.

7. Chip Thickness Ratio and Related Relations

The chip becomes thicker and shorter than the layer removed from the work because of plastic compression.

$$\textbf{Chip thickness ratio, } r = t_1 / t_2$$

$$\textbf{Chip reduction coefficient, } \xi = t_2 / t_1 = 1 / r$$

For orthogonal cutting, the shear angle (φ) relation is:

$$\tan(\varphi) = [r \cos(\alpha)] / [1 - r \sin(\alpha)]$$

where t1 is uncut chip thickness, t2 is chip thickness, α is rake angle and φ is shear angle.

- A smaller chip thickness ratio means greater chip compression.
- Higher positive rake angle usually reduces chip deformation and cutting force.
- Chip thickness can be measured with a micrometer after ensuring that chip curl does not produce a false reading.

8. Chip Breakers

Long continuous chips are difficult and unsafe to handle. A chip breaker forces the chip to curl sharply so that bending stress causes it to fracture into short manageable pieces.

Type	Construction	Remarks
Integral / groove chip breaker	Groove formed or ground on the rake face close to the cutting edge.	Common on indexable inserts and production tools.
Step / obstruction type	A step or raised obstruction causes chip curling and breaking.	Simple and effective over a selected feed range.
Clamp-on chip breaker	Separate plate or clamp is fitted on the tool face.	Adjustable and useful with brazed or conventional tools.

- Chip-breaking action depends on feed, depth of cut, cutting speed, rake angle and chip-breaker geometry.
- A chip breaker designed for heavy feed may not work at very small feed.
- Poorly selected chip breakers can increase force, temperature or tool wear.

DIFFERENT TYPES OF CHIPS

1. Types of Chips

During cutting, the material ahead of the tool is compressed and undergoes plastic deformation. The deformed material flows over the rake face as a chip. The chip type depends on work material, cutting speed, feed, depth of cut, rake angle, friction and lubrication.

Chip type	Favourable / typical conditions	Characteristics and effects
Continuous chips	Ductile work material; high cutting speed; low feed and depth of cut; high back-rake angle; sharp tool and good lubrication.	Long, unbroken chips flow over the tool. They generally give good surface finish, but may entangle, interfere with automated machining and create a safety hazard. A chip breaker is often needed.
Discontinuous chips	Brittle work material; low cutting speed; high feed and depth of cut; low back-rake angle. Vibration and high friction may also encourage segmentation.	Chips separate into short segments. Cutting forces may fluctuate and the machined surface may be rough or uneven.
Continuous chips with built-up edge (BUE)	Ductile material; low cutting speed; high feed and depth of cut; small/low rake angle; high friction and long chip–tool contact.	Work material adheres to the tool tip and gradually forms a built-up edge. It changes the effective cutting-edge geometry, can increase cutting force and cause poor or irregular surface finish when fragments detach.
Serrated / segmented chips	Often associated with high-speed cutting of difficult-to-machine alloys and localized shear.	Saw-tooth-like segments form, with cyclic variations in cutting force and temperature.

1.1 Built-up Edge (BUE): formation and control

A built-up edge consists of workpiece material that gradually deposits and welds near the tool tip during cutting. Under high pressure and temperature, clean metal surfaces at the chip–tool interface may bond locally. The deposited material begins as a small embryo, grows, and may eventually become unstable and break away. Some material can travel with the chip, while some may be carried onto the workpiece surface.

- **Effects:** BUE changes the geometry of the cutting edge, may increase the force required for cutting, and can impair surface finish when it breaks away.
- **Important qualification:** the supplied reference notes that a thin, stable BUE can sometimes protect the rake face and reduce tool wear, even though BUE is generally undesirable.
- **Ways to reduce BUE:** use higher cutting speed, a higher rake angle, and low feed and depth of cut. Good lubrication and a sharp tool can also reduce friction and adhesion.

2. Chip Breakers

Long continuous chips are difficult and potentially dangerous because they can wrap around the tool or workpiece, interfere with cutting and damage the finished surface. A chip breaker makes the chip curl more tightly; the resulting bending stress helps it fracture into short, manageable pieces.

Type	Construction / action	Use / note
Integral or groove type	A groove is formed or ground on the rake face close to the cutting edge. It guides the chip into a tighter curl.	Commonly built into indexable inserts and production cutting tools.
Step / obstruction type	A step or raised obstruction on the rake face forces the chip to bend sharply.	Simple and effective over a selected feed range.
Clamp-on type	A separate plate is clamped to the rake face, obstructing and curling the moving chip.	Can be adjustable; used with conventional tools and some brazed tools.

2.1 Factors affecting chip breaking

- Feed and depth of cut influence chip thickness and stiffness; a chip breaker designed for heavy feed may not work effectively at very low feed.
- Cutting speed, rake angle, chip-breaker shape and the work material also influence the chip curl and breakage.
- A poorly selected chip breaker can increase cutting force, temperature or tool wear.

2.2 Related machining notes from the reference pages

- A large share of the heat generated during cutting is carried away by the chip. The tool therefore receives a smaller share of the heat, although its local temperature can still be high.
- The supplied page notes that feed has more effect on surface roughness than nose radius, and that depth of cut has no effect on surface roughness in the stated context. Treat these as the reference page's simplified exam notes; actual finish can depend on tool geometry, vibration, wear and cutting conditions.

Quick revision

Continuous chip: ductile material + high speed + low feed/depth + high rake. **Discontinuous chip:** brittle material + low speed + high feed/depth + low rake. **BUE:** ductile material + low speed/high friction; reduce with higher speed, higher rake, low feed/depth and good lubrication. **Chip breaker:** curls a long chip sharply so it breaks into short pieces.

These supplementary pages are inserted immediately after the original chip-formation / chip-ratio / chip-breaker material. The original pages are otherwise retained.

9. Forces Acting on a Chip in Orthogonal Cutting

In orthogonal cutting the resultant force can be resolved into components either along the cutting direction, at the tool-chip interface, or along the shear plane.

Symbol	Force	Meaning
Fc	Cutting force	Acts in direction of cutting velocity; largest external force component.
Ft	Thrust force	Acts normal to cutting direction in the orthogonal plane.
F	Friction force	Acts along the rake face opposing chip motion.
N	Normal force	Acts normal to the rake face.
Fs	Shear force	Acts along the shear plane.
Fn	Normal force on shear plane	Acts normal to the shear plane.

Using rake angle α:

$$F = F_c \sin(\alpha) + F_t \cos(\alpha)$$

$$N = F_c \cos(\alpha) - F_t \sin(\alpha)$$

$$F_s = F_c \cos(\varphi) - F_t \sin(\varphi)$$

$$F_n = F_c \sin(\varphi) + F_t \cos(\varphi)$$

$$\text{Coefficient of friction } \mu = F / N = \tan(\beta)$$

where β is the friction angle at the tool-chip interface.

10. Merchant's Circle Diagram and Force Relations

Merchant's circle is a graφcal representation of the force equilibrium in orthogonal cutting. A single resultant force R is resolved into the pairs (Fc, Ft), (F, N) and (Fs, Fn). The circle provides a convenient way to relate cutting force, friction and shear-plane forces.

10.1 Merchant shear-angle relation

Merchant proposed a minimum-energy approach in which the shear angle is related to rake angle α and friction angle β by:

$$\varphi = 45° + \alpha/2 - \beta/2$$

This relation is an idealized metal-cutting model. Real cutting may deviate because deformation is distributed over a shear zone, material properties vary with strain/temperature, and tool-chip contact is complex.

10.2 Useful quantities from the force system

$$\text{Cutting power } P = F_c \times V$$

$$\text{Specific cutting energy } u = P / \text{material removal rate}$$

$$\text{Friction angle } \beta = \tan^{-1}(F/N)$$

$$\text{Shear stress on shear plane } \tau_s = F_s / A_s$$

$$\text{Shear-plane area } A_s = (b \times t_1) / \sin(\varphi)$$

where b is width of cut.

11. Forces on a Single Point Tool in Orthogonal and Oblique Cutting

11.1 Orthogonal cutting

The external force on the single-point tool is generally resolved into cutting force Fc and thrust force Ft. In a turning operation with ideal orthogonal geometry, the radial/side component can be negligible.

11.2 Oblique cutting

In oblique cutting, the force system is three-dimensional. The resultant cutting force is resolved into:

- Tangential or cutting force: acts in the direction of cutting speed and largely determines cutting power.
- Feed or axial force: acts in the feed direction.
- Radial or thrust force: acts normal to the machined cylindrical surface and influences deflection/chatter.

The magnitudes depend on rake angle, inclination angle, feed, depth of cut, work material, cutting speed, tool wear and lubrication.

12. Tool Wear and Tool Failure

12.1 Main forms of tool wear

Wear type	Location / cause	Effect
Flank wear	Wear land on clearance/flank face due to rubbing against machined surface.	Loss of dimensional accuracy, force and temperature rise.
Crater wear	Depression on rake face in chip-flow region.	Weakens cutting edge; severe crater may cause edge collapse.
Notch wear	Localized wear near depth-of-cut line.	Can cause poor finish and sudden edge failure.
Nose wear	Wear at tool nose.	Changes nose radius and dimensional accuracy.

12.2 Mechanisms of tool wear

Mechanism	Description
Abrasion	Hard particles or work-hardened constituents scratch and remove tool material.
Adhesion	Microscopic welding at tool-chip/work interfaces followed by tearing during sliding.
Diffusion	At high temperature, atoms diffuse across the tool-chip interface, weakening the tool material.
Oxidation / chemical wear	Tool reacts chemically with environment or work material at elevated temperature.
Plastic deformation	Cutting edge deforms permanently when hot hardness or compressive strength is insufficient.
Thermal / mechanical fatigue	Repeated temperature or force cycling initiates cracks, especially in interrupted cutting.

12.3 Tool failure modes

- Gradual wear failure: flank or crater wear reaches an unacceptable limit.
- Sudden fracture: cutting edge chips or breaks due to impact, overload or low toughness.
- Plastic deformation: edge loses shape at high temperature/load.
- Thermal cracking: repeated heating/cooling produces cracks, common in interrupted cutting.
- Premature failure from chatter, incorrect geometry, improper clamping or poor tool material selection.

13. Tool Life and Taylor's Tool Life Equation

Tool life is the useful cutting time from a fresh or reground condition until a specified wear or failure criterion is reached. A common criterion is a specified maximum flank-wear land.

13.1 Factors affecting tool life

- Cutting speed: strongest influence in ordinary machining; increasing speed generally reduces tool life sharply.
- Feed and depth of cut: higher values raise cutting force and temperature and generally reduce life.
- Tool material and coating: hot hardness, chemical stability and toughness are critical.
- Work material: hardness, abrasiveness, work hardening and chemical affinity affect wear.
- Tool geometry: rake, clearance, edge preparation and nose radius alter stress and temperature.
- Cutting fluid and delivery method: may reduce friction, temperature and built-up edge.
- Rigidity, vibration and interrupted cutting: impact/chatter reduce tool life.
- Required wear criterion and surface-finish tolerance.

13.2 Taylor's tool-life equation

$$V T^n = C$$

where V is cutting speed, T is tool life, n is the tool-life exponent and C is a constant for a given tool-work combination, tool geometry and cutting conditions.

- On a log-log plot of cutting speed versus tool life, Taylor's basic equation gives a straight line.
- The constant C numerically equals the cutting speed that gives a tool life of one unit of time in the chosen units.
- Values of n and C depend strongly on tool material and work material.

14. Types of Cutting Tool Materials

Tool material	Important properties	Typical use
Carbon tool steel	Low cost, easy to shape and heat treat; loses hardness at modest temperature.	Low-speed hand tools and light cutting.
High-speed steel (HSS)	Good toughness, hot hardness and grindability.	Drills, taps, milling cutters, general-purpose tools.
Cast cobalt alloys	Better hot hardness than HSS but lower toughness.	Special cutting applications; less common today.
Cemented carbides	Very high hardness and wear resistance; much higher cutting speed than HSS.	Turning, milling, production machining.
Coated carbides	Carbide substrate with wear-resistant coatings such as TiN/TiCN/Al2O3.	High-productivity machining, improved crater/flank wear resistance.
Cermets	Ceramic-metal composite, often Ti-based; good wear and finish.	Finishing/semi-finishing steels.
Ceramics	Excellent hot hardness and chemical stability but low toughness.	High-speed finishing of cast iron and hardened materials.
CBN	Very high hardness and thermal stability.	Hardened steels, superalloys, hard turning.
Diamond / PCD	Highest hardness, excellent wear resistance; unsuitable for ferrous materials at high temperature.	Aluminium alloys, copper, composites, non-ferrous abrasive materials.

14.1 Desirable properties of cutting-tool materials

- High hardness and hot hardness.
- High wear resistance.
- Adequate toughness to resist shock and chipping.
- High compressive strength.
- Good thermal conductivity where beneficial.
- Low chemical affinity with work material.
- Resistance to thermal shock and oxidation.
- Ability to retain a sharp, stable cutting edge at working temperature.
- Reasonable cost and ease of manufacture/regrinding where applicable.

15. Metal Working / Cutting Fluids

15.1 Functions of cutting fluids

- Cooling: remove heat from tool, work and chip.
- Lubrication: reduce friction at tool-chip and tool-work interfaces.
- Improve tool life by lowering temperature/friction and suppressing built-up edge.
- Improve surface finish and dimensional accuracy.
- Flush chips away from the cutting zone.
- Prevent rust/corrosion of machine and workpiece when suitable inhibitors are present.
- Reduce welding/seizure in severe cutting operations.

15.2 Qualities of a good cutting fluid

- High cooling capacity where cooling is important.

- Good lubricating and anti-weld properties.
- Chemical stability and resistance to rancidity.
- Non-corrosive and non-staining to work, tool and machine.
- Low toxicity, low odour and acceptable operator safety.
- Low foaming tendency and suitable viscosity.
- Good wetting and penetration.
- Economical, easily filtered and maintainable.

15.3 Types of cutting fluids

Fluid type	Characteristics	Typical application
Water	Excellent cooling but poor lubrication and corrosion risk.	Grinding or special high-cooling applications with additives.
Soluble oils / emulsions	Oil-in-water emulsion; good cooling with moderate lubrication.	General machining.
Straight cutting oils	Mineral/vegetable oils without dilution; good lubrication, lower cooling capacity.	Low-speed severe cutting, threading, broaching.
Synthetic fluids	Water-based chemical solutions without mineral oil.	High cooling, grinding, production machining.
Semi-synthetic fluids	Fine emulsion combining synthetic and oil characteristics.	General CNC and production machining.
Extreme-pressure fluids	Contain additives that form protective films under severe pressure/temperature.	Tapping, broaching, deep drilling, difficult materials.

15.4 Selection of cutting fluids

Machining condition	Selection guideline
High-speed light cutting / grinding	Cooling is dominant; water-based or synthetic fluids are generally preferred.
Low-speed heavy cutting	Lubrication is dominant; straight oil or EP fluid may be preferred.
Aluminium and copper alloys	Use non-staining fluid compatible with the alloy.
Stainless steel / difficult alloys	Good lubricity and EP performance are valuable.
Cast iron	Often machined dry because graφte assists lubrication; fluid may create abrasive sludge.
Carbide at very high speed	Often machined dry or with controlled, continuous coolant; intermittent thermal shock should be avoided.

16. Important Points

- Machining removes excess material as chips by controlled relative motion between tool and work.
- The principal heat sources are primary deformation, tool-chip friction and tool-work flank friction.
- Orthogonal cutting is a two-dimensional idealization; practical machining is mostly oblique.
- ASA, ORS and NRS use different reference planes and angle conventions for tool geometry.
- Chip thickness ratio r = t1/t2; chip compression ratio is its reciprocal.
- Continuous chips generally occur in ductile materials; discontinuous chips are common in brittle materials.
- Merchant's circle relates cutting/thrust forces to friction and shear-plane forces.
- Flank wear affects accuracy; crater wear develops on the rake face in the chip-contact region.
- Cutting speed has a very strong influence on tool life.
- Taylor's basic tool-life relation is V T^n = C.
- Cutting-tool materials are selected by balancing hardness, hot hardness, wear resistance and toughness.
- Cutting fluids perform cooling, lubrication, chip flushing and corrosion-control functions.

Reference

Hajra Choudhury, S. K. and Hajra Choudhury, A. K., Elements of Workshop Technology, Vol. II.